User Manual

ADP Hunter™ Plus Assay

For the evaluation of kinase activity and inhibition through fluorometric measurement of ADP accumulation

For Assay Kits:

90-0083: 2-Plate Kit

90-0083M: 6-Plate Kit

90-0083L: 25-Plate Kit

90-0083XL: 125-Plate Kit

Document Number 70-153 Revision 9

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Overview

The ADP Hunter™ Plus assay is a biochemical assay for the evaluation of kinase activity and the screening and profiling of kinase inhibitors directly through measurement of ADP accumulation. ADP Hunter Plus is specifically designed for high throughput screening of kinase inhibitors and is compatible with both peptide substrates and whole proteins. The assay has been designed for research use in 384-well microplates but can also be run in additional microplate formats.

Assay Principle

ADP Hunter Plus Assay is a homogeneous assay for the measurement of ADP, a universal product of kinase activity. This method follows the product of the reaction and is a generic, non-radioactive method which offers a convenient gain-of-signal assay format in 90 minutes. In the ADP Hunter Plus assay a coupled enzyme reaction system is used to generate hydrogen peroxide from ADP. Hydrogen peroxide, when combined with ADHP (fluorescent dye precursor) in the presence of peroxidase generates the fluorescently active resorufin dye. To allow for automation, a Stop Solution is also provided for added signal and background stabilization. See Figure 1 below.

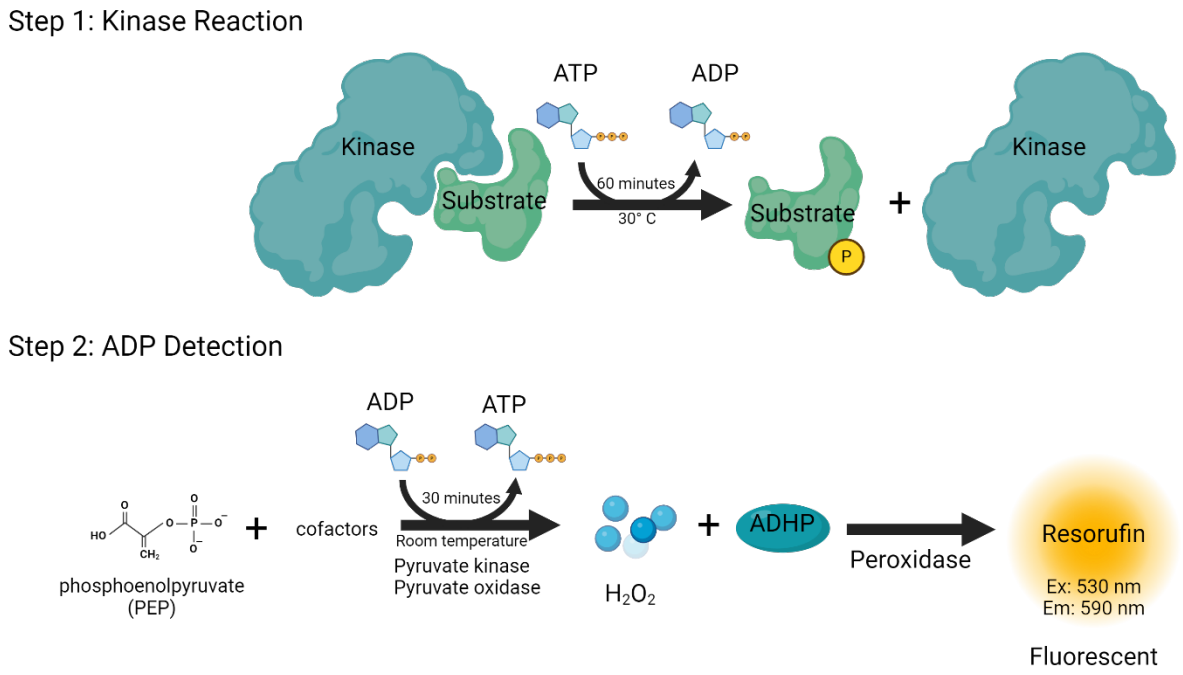

Figure 1**. Kinase Enzyme Activity Assay Principle**

Assay Workflow

Please read the entire protocol completely before running the assay. Successful results depend on performing these steps correctly. The Assay Procedure sections, and the Quick Start Guide contain detailed information about how to run the assays.

Inhibitor Determination

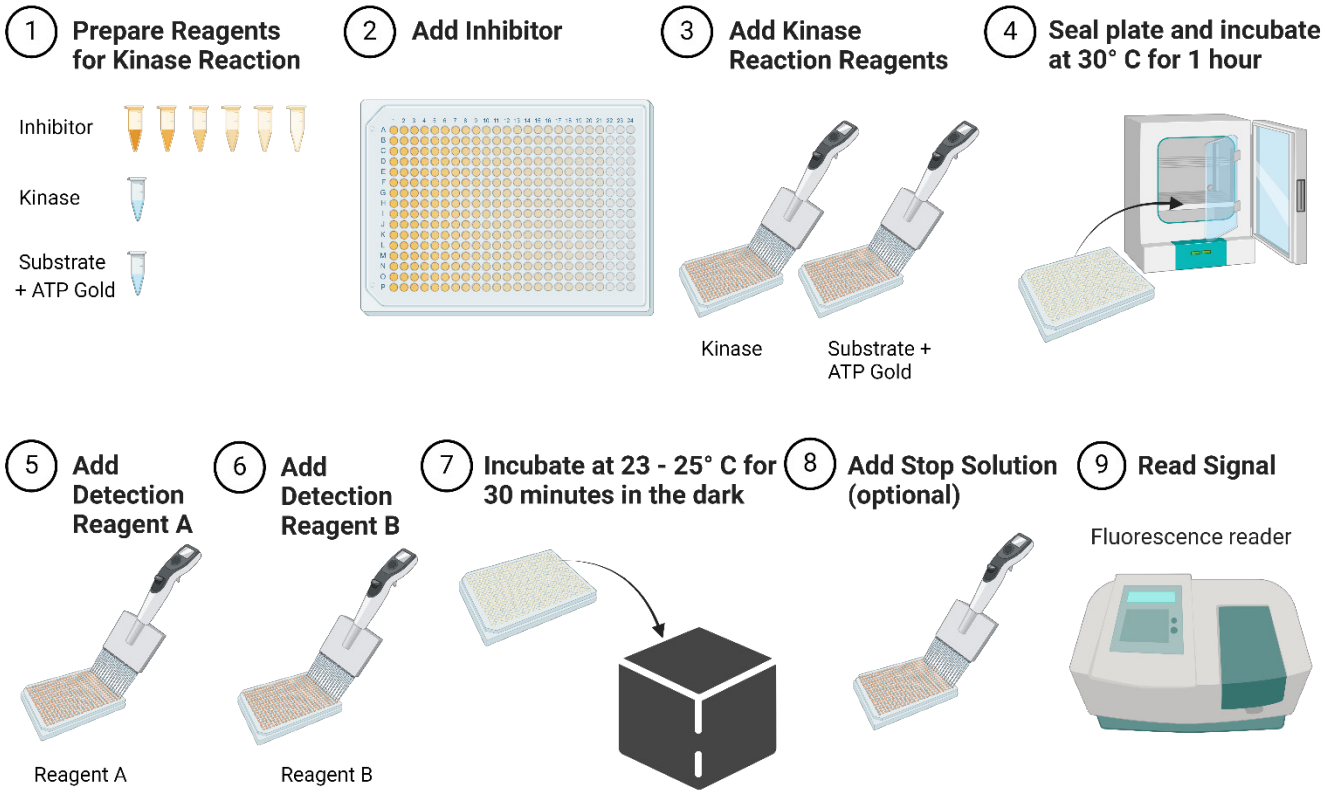

Figure 2. Kinase Reaction Inhibitor Determination Assay Workflow

Screen and profile kinase inhibitors directly through measurement of ADP accumulation by performing the following steps (Figure 2).

ADP Standard Curve

The assay workflows for generating an ADP Standard Curve and performing a Kinase Reaction Procedure for inhibitor determination are described in this section.

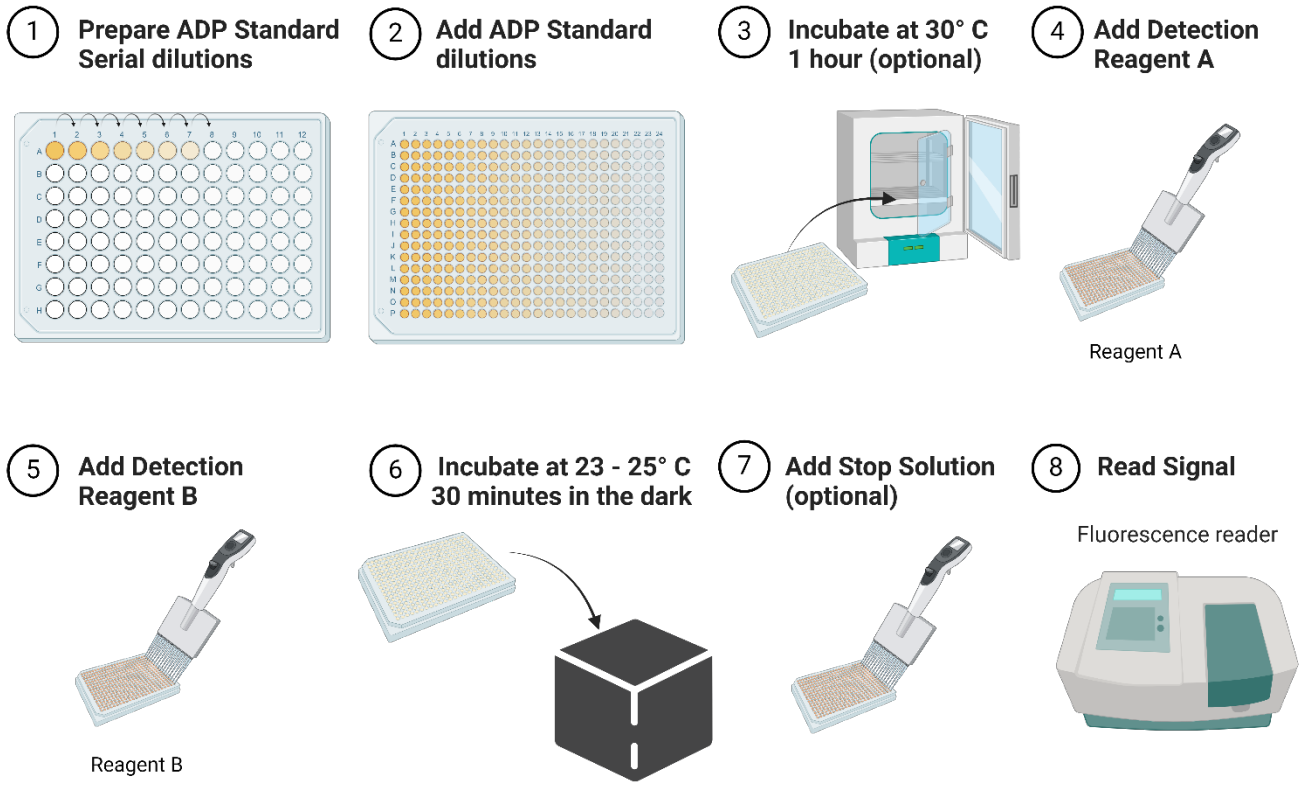

Figure 3. ADP Standard Curve Assay Workflow

Figure 3 illustrates the assay steps required for the generation of the ADP standard curve. For reagent volumes and preparation instructions, refer to the “Detailed Assay Protocol” section of this manual.

Materials Provided

Catalog Number	90-0083	90-0083M	90-0083L	90-0083XL
384-well (full volume), No. of data points	~800	~2,400	~10,000	~50,000
Kit Components				
ADP Hunter Plus Reagent A[1]	8 mL	3 x 8 mL	100 mL	500 mL
ADP Hunter Plus Reagent B[1]	16 mL	3 x 16 mL	200 mL	1000 mL
ADP Hunter Plus Stop Solution	4 mL	3 x 4 mL	50 mL	250 mL
ADP Hunter Plus ADP Standard (225 µM)	2 mL	3 x 2 mL	25 mL	25 mL
ADP Hunter Plus Assay Buffer	50 mL	3 x 50 mL	625 mL	2 x 625 mL

[1] Reagents A and B upon thawing may show some particulate matter. This does not affect the performance of the assay.

Table 1. Materials Provided

Note: The ADP Hunter™ Plus Assay Buffer is used to prepare the substrate, the kinase and ATP Gold. The assay buffer is pH 7.4 containing 15 mM HEPES, 20 mM NaCl, 1 mM EGTA, 0.02% Tween 20, 10 mM $MgCl_2$, and 0.1 mg/mL BGG (bovine-γ-globulins).

Storage and Stability

Upon arrival, store reagents at -20° C or -70° C. Thaw reagents at room temperature before use, and after thawing, store reagents for up to 1 week at 2-8° C. For longer term storage, aliquots of all the components must be re-frozen at -20° C or -70° C. The ADP Hunter™ Plus Kit is stable until the expiration date indicated on the box outer label when stored at recommended storage temperatures.

Handling Recommendations

ADP Hunter Plus Kit and ATP Gold: Equilibrate reagents to ambient room temperature before use. Reagents can be thawed up to 3 times with no effect on performance.

Additional Materials Required

The following materials are required.

Equipment	Materials
Fluorescence intensity reader	Pipettes and pipette tips
Filter wavelengths required for Resorufin: Peak Excitation wavelength: 530 nm Peak Emission wavelength: 590 nm	Kinase enzyme (discoverx.com/target-class/kinase/) Kinase substrate 96-well plate for compound dilution Cat. No. 92-0011 ATP Gold (25 mM) Cat. No. 90-0099 384-well Non-Binding Black, flat bottom plate (e.g., Greiner 791800) Plate seal

Table 2. Materials Required

Guidelines for Use

The assay consists of three steps, step 1 is the kinase reaction, step 2 is the addition of the detection reagents, and step 3 is the addition of the assay stop solution (optional).

Kinase Reaction: The kinase reaction includes a purified active kinase, a substrate (peptide, protein, or lipid), ATP, compounds, all in an appropriate kinase reaction buffer.

Kinase Reaction Buffer: An assay buffer is provided which may be used as the kinase reaction buffer. If using a different kinase reaction buffer, it is imperative to dilute the ADP standard in your buffer and test for any interference with the detection reagents. The assay is optimal in 6.0-8.0 pH range and can tolerate the following concentrations of commonly used components of the kinase reaction. The concentrations here refer to final kinase reaction concentrations:

ATP	≤ 200 mM	EDTA	≤ 10 mM	NaCl	≤ 150 mM
BGG	≤ 2 mg/mL	EGTA	≤ 10 mM	$MgCl_2$	≤ 10 mM
DMSO	≤ 5%	Glycerol	≤ 5%	Sodium Orthovanadate	≤ 2 mM
DTT	≤ 2 mM	HEPES	≤ 15 mM	β-mercaptoethanol	≤ 0.01%

Table 3. Kinase reaction buffer tolerance specifications

ADP Hunter™ Plus Reagent A: Ready to use, no preparation needed.

ADP Hunter Plus Reagent B: Ready to use, no preparation needed.

ADP Hunter Plus Stop Solution: Ready to use, no preparation needed.

Controls: A control without kinase should be included in each experiment and signal from this control may be used for correction of background signal, this control should contain ATP, substrate, and compound or vehicle.

Tips for optimal performance:

- Use the highest quality ATP in performing kinase reactions. Some sources of ATP contain high amounts of contaminating ADP that can significantly increase assay background. We recommend using ATP Gold (DiscoverX™, Cat. # 90-0099). ATP powder should be stored desiccated at -20° C. Ideally fresh ATP solutions should be used and made up in 100 mM HEPES, pH 7.4. Alternatively stock solution can be aliquoted and stored frozen at -70° C. Do not freeze-thaw ATP stock solution.
- Reagent A must be added before Reagent B for optimal assay performance.
- If needed, Stop Solution must be added last after the 30-minute incubation of Reagent A and B. Do not combine with either Reagent A or Reagent B.
- The Stop Solution stabilizes signal and background.

Protocol Schematic

Tip: *Use this sheet to note your assay specific conditions. Post it on your bench to use as a quick reference guide.*

Assay Name: _____ Date: _____

Product Details: _____

Quick-start Procedure: In a 384-well microtiter plate, perform the following steps.

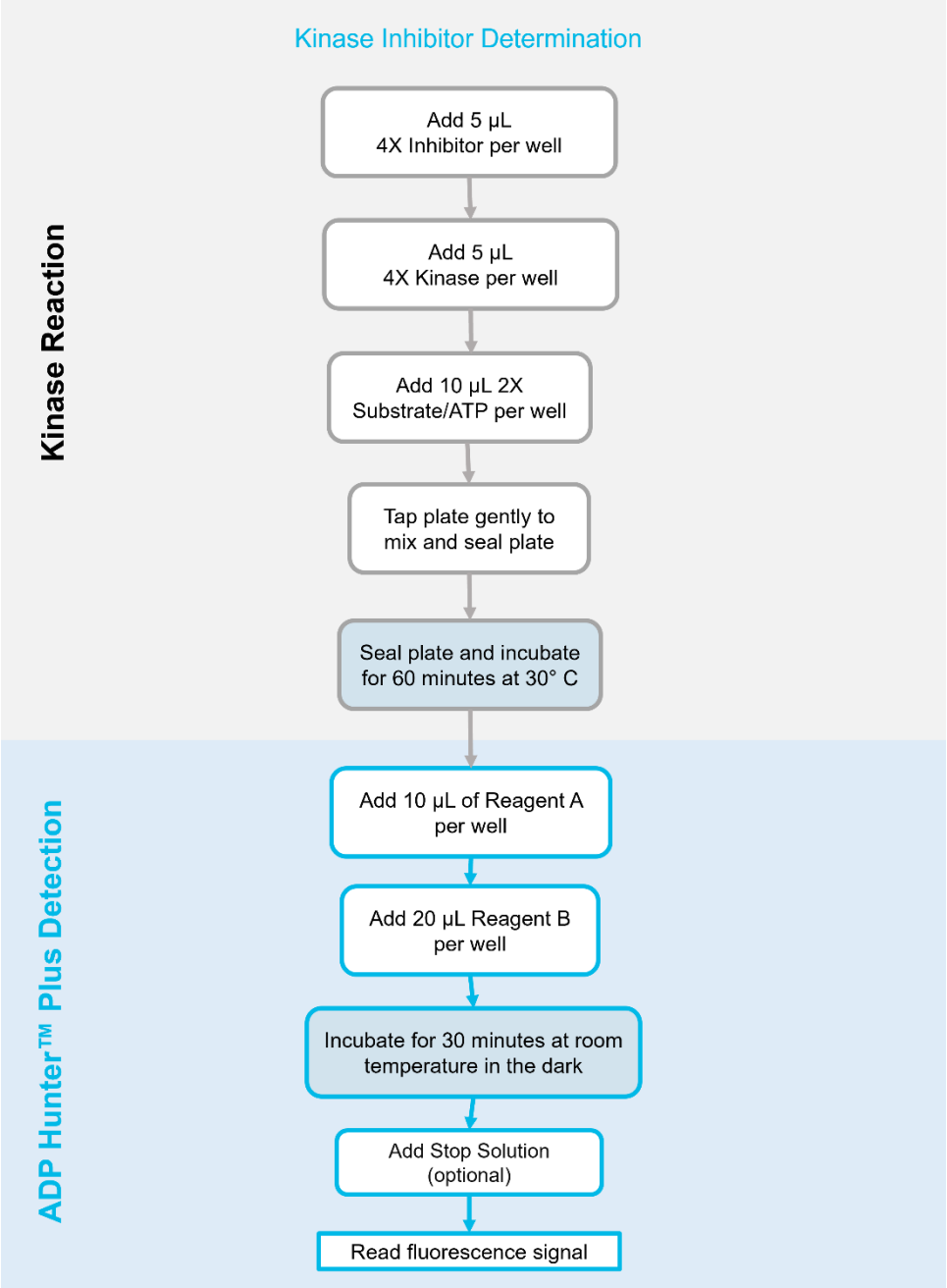

Figure 4. Quick-start Procedure

Detailed Assay Protocol (Standard Curve)

ADP Standard Preparation

Prepare 3-fold serial dilutions of the ADP Standard in the Assay Buffer provided in the kit. Add 100 µL of Assay Buffer per well in 8 wells of a 96-well plate. Add 50 µL of the Standard to the first well. Then transfer 50 µL to the next well. Serially dilute 5 more times and do not add any diluted Standard to the last well. Use the assay buffer in the last well as the zero ADP control. The ADP Standard concentrations range from 0.1 µM to 75 µM in the kinase reaction.

Note: 40 µL per concentration is required for one standard curve (7 points curve plus zero control, duplicate replicas per concentration).

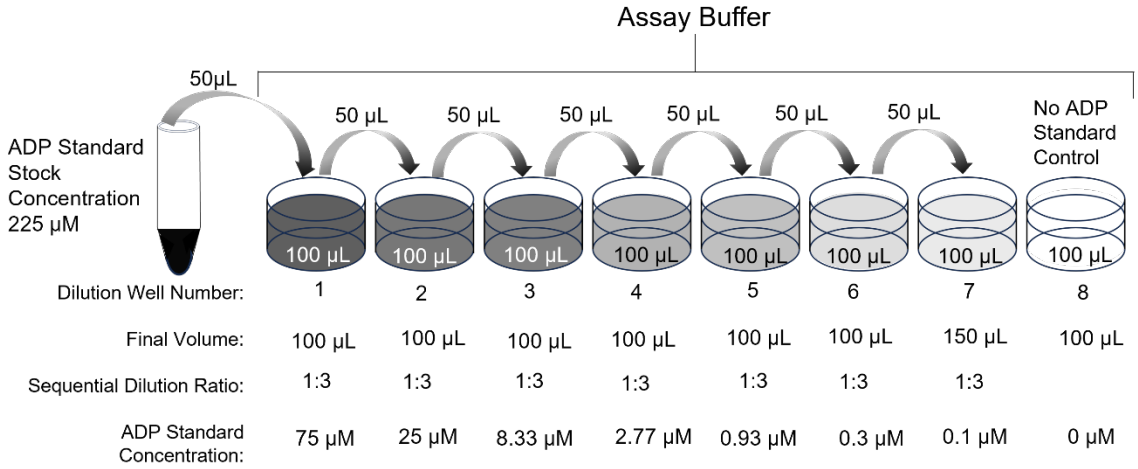

Figure 5. ADP Standard Serial Dilution

ADP Standard Addition, Detection and Plate Reading

The table below outlines the volumes and procedure for the ADP standard curve for a full volume 384-well plate format.

Assay Step	ADP standard
Step 1: Standard Dilutions	20 µL ADP Standard dilutions
Incubate 1 hour at 30° C (not necessary)	
Step 2: ADP Detection	Add 10 µL Reagent A Then Add 20 µL Reagent B
Incubate 30 minutes at room temperature (protect from light)	
Step 3: Stop Solution (optional)	Add 5 µL Stop Solution
Read Fluorescence Intensity signal: Resorufin Excitation wavelength – 530 nm Resorufin Emission wavelength – 590 nm	
Note: The signal may be measured up to 1 hour after addition of Stop Solution. The addition of Stop Solution will result in a 10% reduction in assay window.	

Table 4. ADP Standard Curve Assay Steps

Detailed Assay Protocol (Kinase Reaction Procedure for Inhibitor Determination)

Inhibitor compounds, Kinase, substrate, and ATP Gold must be diluted in the ADP Hunter™ Plus Assay Buffer. Inhibitor compounds are prepared at a 4X concentration and 5 µL added per well. Kinase is prepared at a 4X concentration and 5 µL added per well. A master mix of substrate and ATP is prepared at 2X of the final assay concentration and added at 10 µL per well to initiate the kinase reaction. The total kinase reaction volume is 20 µL.

Inhibitor Preparation

1. Prepare a 4X master mix of inhibitor in ADP Hunter Plus Assay Buffer
   - For single point determination, 10 µM is a typical test concentration. Include several wells for the vehicle control as this defines activity in the absence of kinase inhibition.
2. Add **5 µL 4X** Inhibitor into each test well of the assay plate.
3. Cover the plate to avoid evaporation and proceed to the next step.

Kinase Preparation

1. Prepare a 4X master mix of kinase in ADP Hunter Plus Assay Buffer.
2. Add **5 µL 4X** kinase to all wells that contain inhibitor or vehicle control.
3. After the kinase is added, the plate can be tapped gently to mix with compound.
4. Cover the plate to avoid evaporation and proceed to next step.

Substrate and ATP Gold Preparation

1. Prepare a 2X master mix of Substrate and ATP Gold in ADP Hunter Plus Assay Buffer.
2. Add **10 µL 2X** Substrate/ATP master mix into each well of the assay plate.
3. After the mix is added, the plate can be tapped gently to mix the kinase reaction.
4. Seal the plate with the provided plate seal.
5. Incubate the plate for 60 minutes at 30° C.

Note: The reaction is initiated with the addition of ATP.

Inhibitor, Kinase, Substrate and ATP Gold Additions, Detection and Plate Reading

The table below indicates assay volumes and procedure for full volume 384-well format.

Run appropriate controls with all experiments.

Assay step	Reagent volumes per well
Step 1: Kinase Reaction	5 µL of inhibitor preparation 5 µL of Kinase preparation 10 µL of Substrate and ATP Gold preparation
Seal plate and incubate 1 hour at 30° C	
Step 2: ADP Detection	Add 10 µL Reagent A Then Add 20 µL Reagent B
Incubate 30 minutes at room temperature (protect from light)	
Step 3: Stop Solution (optional)	Add 5 µL Stop Solution
Read Fluorescence Intensity signal: Resorufin Excitation wavelength – 530 nm Resorufin Emission wavelength – 590 nm	
Note: The signal may be measured up to 1 hour after addition of Stop Solution. The addition of Stop Solution will result in a 10% reduction in assay window.	

Table 5. Kinase reaction procedure for Inhibitor determination assay steps

Representative Data

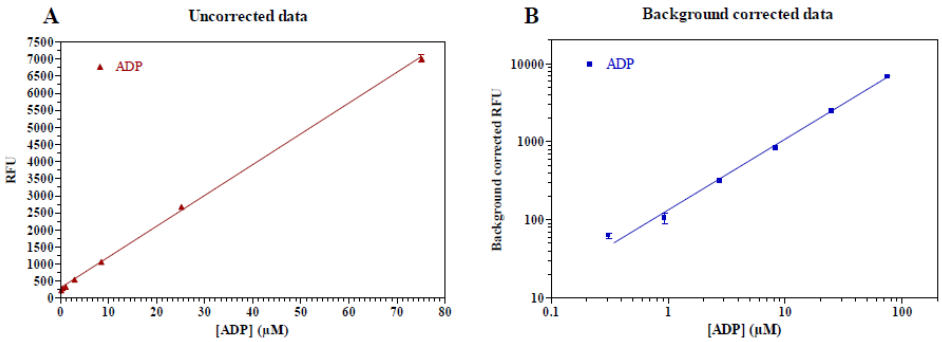

Figure 6. ADP Hunter™ Plus standard curve

The ADP Hunter Plus standard curve is illustrated above. (A. Uncorrected data. B. Data after background correction [0 µM ADP]). The dynamic range of the assay is 0.6 µM to 75 µM ADP. The data is shown using a linear fit (r^2 = 0.998).

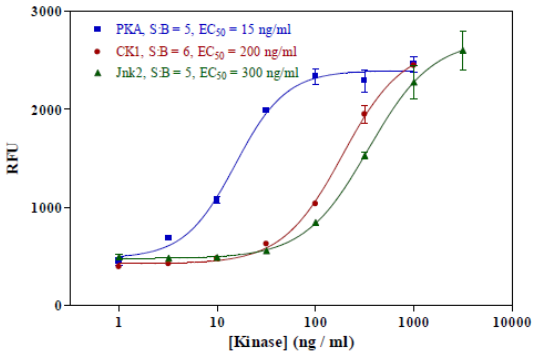

Figure 7. Kinase activity curves

Kinase activity curves for PKA, CK1 and Jnk2 in the presence of 25 µM ATP. Kinase reactions were performed at 30° C for 60 minutes.

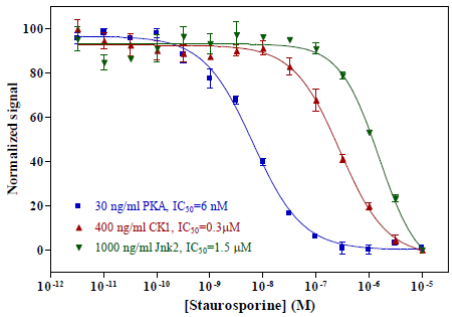

Figure 8. Staurosporine inhibition of kinase activity

Staurosporine inhibition of kinase activity in the presence of 25 µM ATP. Kinase reactions were performed at 30° C for 60 minutes. Data shown is normalized as percentage signal change.

Document Revision History

Revision Number	Date Released	Revision Details
9	July 2024	Kit Component, brand format and other updates for consistency and clarity.

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